



Vaccine efficacy at a point in time

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SUMMARY

Vaccine trials are generally designed to assess efficacy on clinical disease. The vaccine effect on infection, while important both as a proxy for transmission and to describe a vaccine's entire effects, requires frequent (e.g., twice a week) longitudinal sampling to capture all infections. Such sampling may not always be feasible. A logistically easy approach is to collect a sample to test for infection at a regularly scheduled visit. Such point or cross-sectional sampling does not permit estimation of classic vaccine efficacy on infection, as long duration infections are sampled with higher probability. Building on work by Rinta-Kokko and others (2009) and Lipsitch and Kahn (2021), we evaluate proxies of the vaccine effect on transmission at a point in time; the vaccine efficacy on prevalent infection and on prevalent viral load, VE_{PI} and VE_{PVL} , respectively. Longer infections with higher viral loads should have more transmission potential and prevalent vaccine efficacy naturally captures this aspect. We demonstrate how these parameters obtain from an underlying proportional hazards model for infection and allow for waning efficacy on infection, duration, and viral load. We estimate these parameters based on regression models with either repeated cross-sectional sampling or frequent longitudinal sampling. We evaluate the methods by simulation and analyze a phase III vaccine trial with polymerase chain reaction (PCR) cross-sectional sampling for subclinical infection.

Keywords: COVID-19; Cross-section; Infectious disease; Prevalent study.

1. INTRODUCTION

Many vaccine trials do not directly assess vaccine efficacy on infection as there is no sampling of asymptomatic volunteers to detect the presence of the pathogen. Yet the vaccine effect on infection is important to understand, both for its potential impact on transmission and to better characterize the vaccine effect on individuals. Regularly sampling trial volunteers is difficult due to the additional logistical burden imposed on volunteers and study personnel. Volunteers do periodically come in for exams and may come in for special visits (e.g., once the vaccine has been shown to be efficacious, a visit for all participants may be scheduled so that participants in the placebo arm may get the vaccine). At such visits, a sample to detect infection by, for example, a polymerase chain reaction (PCR) test, could be collected to assess the proportion of infections in the two arms. The vaccine efficacy on the reduction in PCR positive tests at this point in time could be calculated. Yet what does it measure?

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A vaccine has myriad effects, and multiple definitions of vaccine efficacy have been proposed: VE on disease, infection (susceptibility), and transmission, acting on both individuals and the population (see [Halloran and others, 1999](#)). Cross-sectional sample estimates do not accurately recover the true vaccine efficacy on infection for a person, that is, VE_I , as longer duration infections are over-represented from a single cross-sectional sample. [Rinta-Kokko and others \(2009\)](#) introduced the vaccine efficacy on pneumococcal carriage which we call the vaccine efficacy on prevalent infection or VE_{PI} (see also [Lipsitch and Kahn, 2021](#)). This should be a better proxy for the effect of the vaccine on transmission than VE_I , as it reflects that longer infections have more transmission potential. We also introduce the vaccine efficacy on prevalent viral load or VE_{PVL} . This is the proportion reduction in the amount of virus, generally after log transformation, in a vaccinated person at a point in time. Since more viruses should increase transmissibility, this may be an even better proxy for transmission. In HIV, for example, population-level viral load is used as a metric to evaluate the effectiveness of population-level interventions ([Wood and others, 2009](#)). A feature of these metrics is that they will be larger than VE_I unless the vaccine prolongs or increases the viral load trajectory.

In this work, we motivate and develop these three types of vaccine efficacy. We apply a proportional hazards model for the instantaneous risk of infection over time and show how VE_I , VE_{PI} , and VE_{PVL} are expressible as functions of the parameters of this model which is a form of a mark specific hazard (see [Gilbert and others, 2004](#)), and this model has been previously used to estimate vaccine efficacy on the number of founding viruses of an infection (see [Follmann and Huang, 2015](#)). The model allows us to demonstrate how VE_{PI} and VE_{PVL} can be estimated with frequent longitudinal sampling (e.g., daily or twice a week), and how VE_I can be estimated using cross-sectional data, provided the vaccine to placebo ratio of mean infection duration, or viral load is known. This generalizes results between incidence, duration, and prevalence to viral loads (see [Freeman and Hutchison, 1980](#); [Keiding, 1991](#)). We show how to combine multiple cross-sectional samples by formulating regression models for infection and on viral load given infection. These models readily incorporate covariates and functions of time since vaccination, s , which provide adjusted estimates of $VE_{PI}(s)$ and $VE_{PVL}(s)$. We also discuss how to combine cross-sectional and proportional hazards estimates. We evaluate the estimators via simulation and analyze the day 28 cross-sectional sample data from the phase III Moderna Coronavirus disease of 2019 (COVID-19) vaccine trial ([Moderna, 2020](#)).

2. MOTIVATION

Vaccine efficacy for infection (susceptibility) typically requires frequent evaluation. For example, for COVID-19 vaccine trials, twice weekly PCR testing might be required to ensure nearly all infection events are captured. For large studies with a rare disease, a good estimate of the vaccine efficacy on infection is

$$\widehat{VE}_I = 1 - \frac{\bar{X}_1}{\bar{X}_0}, \quad (2.1)$$

where $\bar{X}_Z$ is the observed proportion of infections recorded in the volunteers from arm Z measured over a common period of time (see [Figure 1](#)). In practice, this might be all volunteers, baseline seronegative volunteers, or baseline seronegative volunteers without prior symptomatic disease, depending on the question.

Frequent sampling for infection can be burdensome and expensive. A logistically easy approach is to sample subjects at a single point in time such as a crossover or serology visit. Define $\bar{Y}_Z$ as the proportion of tested individuals in arm Z with an infection at such a visit. We can form a simple estimate of vaccine

efficacy for infection at this point in time as

$$\widehat{\text{VE}}_{\text{PI}} = 1 - \frac{\bar{Y}_1}{\bar{Y}_0}, \quad (2.2)$$

which we call the vaccine efficacy for prevalent infection or VE_{PI} . This approach was introduced by [Rinta-Kokko and others \(2009\)](#) to describe the vaccine effect on pneumococcal carriage, though their metric was the odds ratio (also see [Thompson and others, 1998](#)). This differs from $\widehat{\text{VE}}_I$. If the vaccine tends to make the duration of infection shorter, then fewer vaccine infections will be collected when sampled at a single point in time as shown in Figure 1. As suggested by [Rinta-Kokko and others \(2009\)](#), and [Lipsitch and Kahn \(2021\)](#), $\widehat{\text{VE}}_{\text{PI}}$ may be a better measure of transmission risk as it reflects the reduction in the number of infected individuals in the community. In contrast, $\widehat{\text{VE}}_I$ measures the effect of vaccination on an individual's risk of infection regardless of duration.

If the cross-sectional sample includes a measure of viral load, we can create an even better proxy for the vaccine effect on transmission. To motivate this estimate, suppose the vaccine has no effect on infection, nor duration, but reduced the viral load during infection by 90%, compared to placebo. Then on any given day, there would be 90% less virus in the vaccine volunteers compared to placebo volunteers. Presumably, this would translate into a substantially reduced risk of onward transmission. We thus form a simple estimate of the vaccine efficacy on prevalent viral load as the proportion reduction in total viral load at a point in time,

$$\widehat{\text{VE}}_{\text{PVL}} = 1 - \frac{V_{1+}/n_1}{V_{0+}/n_0}, \quad (2.3)$$

where V_{Z+} is the sum of viral loads over all sampled volunteers on arm Z and n_Z the number sampled. In assessment of HIV control measures, the change in population-level viral load is used and VE_{PVL} is analogous for the vaccine effect on transmission potential ([Wood and others, 2009](#)). The intuition motivating VE_{PI} and VE_{PVL} is illustrated in Figure 2.

Let Y_{Z+} be the number of infections on arm Z from the cross-sectional sample. Because $V_{Z+}/n_Z = (Y_{Z+}/n_Z) \times (V_{Z+}/Y_{Z+}) = \bar{Y}_Z \times \bar{V}_Z$, (2.3) can be written as

$$\widehat{\text{VE}}_{\text{PVL}} = 1 - \frac{\bar{Y}_1}{\bar{Y}_0} \times \frac{\bar{V}_1}{\bar{V}_0}. \quad (2.4)$$

Thus, $\widehat{\text{VE}}_{\text{PVL}}$ blends a vaccine effect on the probability a person is infected on a given day times an effect on the mean viral load among the infected. In the next section, we develop an infection process model and formally define time-varying $\text{VE}_{\text{PI}}(s)$, $\text{VE}_{\text{PVL}}(s)$, and $\text{VE}_I(s)$ in terms of the parameters of this single process. This model explicitly defines these parameters in a manner so that we may create estimators using real-world data where point sampling is done at multiple times and vaccine efficacy can wane with s , the time since vaccination.

3. THEORETICAL DEVELOPMENT

3.1. Constant vaccine efficacy

VE_I may be parameterized in several ways under different models, for example as one minus a ratio of either incidence rates, cumulative incidence rates, or hazard rates (see e.g., [Halloran and others, 1999](#), Table 1). $\widehat{\text{VE}}_I$ is often a reasonable estimator under any of those parameterizations. Here we parameterize

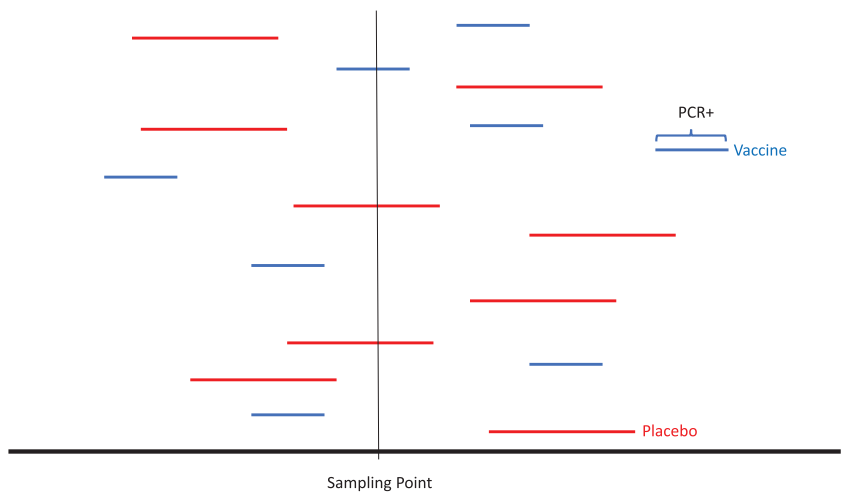


Fig. 1. Two different estimates of vaccine efficacy for infection. The vaccine efficacy for prevalent infection or $\widehat{VE}_{PI} = 1 - 1/2 = 0.50$. The traditional vaccine efficacy counts all infections and is $\widehat{VE}_I = 1 - 8/9 = 0.11$.

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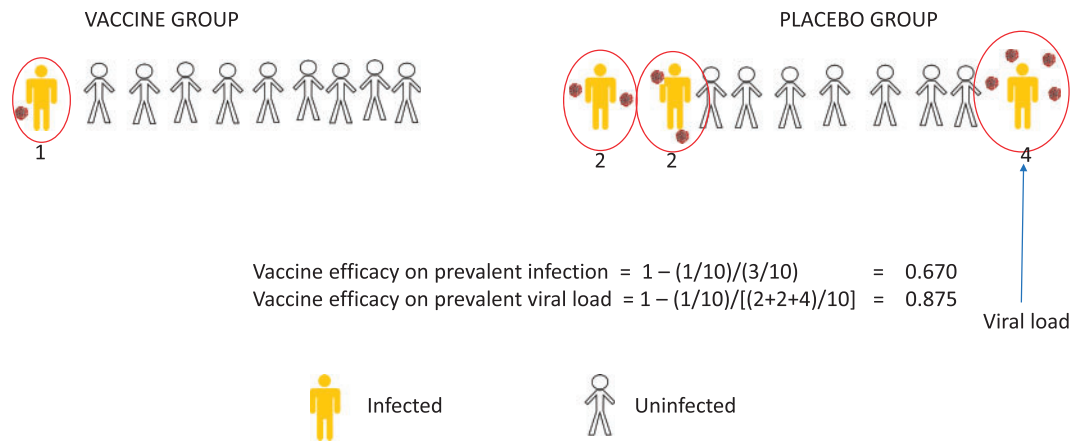


Fig. 2. Vaccine efficacy estimates at a point in time. The placebo group has more infected and they have higher viral loads. The placebo group should be more infectious both when we contrast total infected and especially when we contrast total viral load. Two metrics reflect this idea. The vaccine efficacy for prevalent infection is estimated as $\widehat{VE}_{PI} = 0.67$. The vaccine efficacy for prevalent viral load is estimated as $\widehat{VE}_{PVL} = 0.875$.

it using a proportional hazards model. Let T be the number of days from injection, whether vaccine or placebo, to the start of infection. We assume a discrete time proportional hazards model where the hazard for infection is given by

$$\lambda(s) = \lambda_0(s) \exp(Z\theta), \tag{3.5}$$

where Z is the vaccine indicator, $s = 1, 2, \dots$ the days since vaccination, and $\lambda_0(s)$ an unspecified baseline hazard.

The hazard function representation can be decomposed as $\lambda_0(s) = \omega(s)P(A = 1|Z = 0)$, and $\theta = \log\{P(A = 1|Z = 1)/P(A = 1|Z = 0)\}$, where A is the indicator of acquisition of infection *given* exposure, and $\omega(s)$ is the exposure process common to both groups, as shown in [Rhodes and others \(1996\)](#), [Halloran and others \(2010\)](#), and [Follmann and Huang \(2015\)](#). We define vaccine efficacy against infection as $VE_I = 1 - \exp(\theta)$.

Based on this model we can define the parameters VE_{PI} and VE_{PVL} . For illustration, suppose infections have duration D at most 2 days with $P(D = d|Z) = p_Z(d)$ for $d = 1, 2$ in arm $Z = 0, 1$. A person infected on day $s - 1$, with a duration of 2 days will be detected on day s , and everyone infected on day s will be detected. Thus, the probability an infection is detected on day s is approximately

$$\begin{aligned} E_Z\{Y(s)\} &\approx \lambda_0(s-1) \exp(Z\theta) p_Z(2) + \lambda_0(s) \exp(Z\theta) \{p_Z(2) + p_Z(1)\} \\ &\approx \lambda_0(s) \exp(Z\theta) \{2p_Z(2) + p_Z(1)\} \\ &= \lambda_0(s) \exp(Z\theta) \Delta_Z, \end{aligned} \quad (3.6)$$

where $Y(s)$ is the indicator of infection detection at day s postvaccination, and Δ_Z is the expected duration of infection in arm Z . The first approximation is because $\lambda_0()$ is close to 0 (and $P(T > s)$ is close to 1), and the second assumes $\lambda_0(s-1) = \lambda_0(s)$, that is, that the placebo infection risk is constant just prior to s .

Given (3.6), we can deduce that the ratio of the probability of an active infection from someone on vaccine divided by the probability of an active infection from someone on placebo at day s is approximately

$$\frac{\lambda_0(s) \exp(\theta) \Delta_1}{\lambda_0(s) \Delta_0} = \exp(\theta) \frac{\Delta_1}{\Delta_0}.$$

This well-known result relates prevalence to incidence and duration (see e.g., [Freeman and Hutchison, 1980](#); [Keiding, 1991](#)). Thus, $1 - E_1\{Y(s)\}/E_0\{Y(s)\}$ is approximately independent of s , so we define a time-independent vaccine efficacy for prevalent infection as

$$VE_{PI} = 1 - \exp(\theta) \frac{\Delta_1}{\Delta_0}.$$

Prevalent viral load is more complicated. Let $V(s, b, d)$ be the viral load on day s postvaccination and day b postinfection for an individual whose infection has duration d . It follows that the infection started on day $T = s - b + 1$ postvaccination. Let its expectation be $v_Z(b, d) = E(V(s, b, d)|T = s - b + 1, Z)$ for someone in arm Z given the infection began at $T = s - b + 1$. In this section, we assume the mean viral load only depends on the time since infection, b , and the duration, d , but not on the time since vaccination, s . We have

$$\begin{aligned} E_D[\{V(s, B, D)|D, Z\}] &\approx \lambda_0(s-1) \exp(Z\theta) p_Z(2) v_Z(2, 2) + \lambda_0(s) \exp(Z\theta) \{p_Z(2) v_Z(1, 2) + p_Z(1) v_Z(1, 1)\} \\ &\approx \lambda_0(s) \exp(Z\theta) [p_Z(2) \{v_Z(2, 2) + v_Z(1, 2)\} + p_Z(1) v_Z(1, 1)] \\ &= \lambda_0(s) \exp(Z\theta) \mu_Z, \end{aligned}$$

where μ_Z is the mean area under the viral load curve for those who are infected. Note that the mean viral load on a day randomly chosen by length-biased sampling, say v_Z , selects mean viral loads

$v_Z(1, 1), v_Z(1, 2), v_Z(2, 2)$ with probabilities proportional to $p_Z(1), p_Z(2), p_Z(2)$, respectively. That is,

$$v_Z = \frac{p_Z(1)v_Z(1, 1) + p_Z(2)v_Z(1, 2) + p_Z(2)v_Z(2, 2)}{p_Z(1) + p_Z(2) + p_Z(2)} = \frac{\mu_Z}{\Delta_Z}.$$

So, $\mu_Z = \Delta_Z v_Z$. We define vaccine efficacy on prevalent viral load as

$$\text{VE}_{\text{PVL}} = 1 - \exp(\theta) \frac{\mu_1}{\mu_0} = 1 - \exp(\theta) \frac{\Delta_1 v_1}{\Delta_0 v_0},$$

a blending of the infection risk and the mean viral load as in (2.4).

With this machinery, other proxies for transmission can be formed. For example, if transmission were unlikely to occur unless the viral load exceeded a threshold, say five logs, then we could redefine viral load as $V^* = I(V > 5)$, where $I()$ is the indicator function. Or if there were a known function say $P(V)$ that provided the probability of transmission given a viral load of V , we could use the function $P(V)$ instead of V in the above development.

The above arguments generalize to durations greater than 2 but require constant vaccine efficacy and constant attack rate for an interval M days prior to sampling at s , where M is the maximum duration of infection. We next relax the requirement of time-constant vaccine efficacy.

3.2. Time-varying vaccine efficacy

To allow for time-varying vaccine efficacy, the development for the placebo group is exactly the same. For the vaccine group we need to allow for time-varying vaccine effects on the acquisition of infection, the duration of infection, and the distribution of viral loads. To allow time-varying effects we set $\text{VE}_I(s) = 1 - \exp\{\theta(s)\}$, where s is time since vaccination and define $p_1(d, t)$ as the probability an infection acquired at day t postvaccination that has a duration of d days. We also define $v_1(b, D, T) = E(V(s, b, D)|D, T = s - b + 1, Z = 1)$ which is the expected viral load on day b postinfection for a person in the vaccine arm infected at time T postvaccination (i.e., expected viral load at time $s = T + b - 1$ after vaccination), who has duration of length $D \leq M$ —the maximum duration.

Under this more general structure, the probability of detecting an infection on day s depends on the daily infection risks, approximately proportional to $\lambda_0(s - M + 1) \exp\{Z\theta(s - M + 1)\}, \dots, \lambda_0(s) \exp\{Z\theta(s)\}$, and the infection duration distributions $p_1(d, s - M + 1), \dots, p_1(d, s)$ in a complex way. In the [Supplementary material](#) available at *Biostatistics* online, we show that under the assumption of a constant placebo attack rate for the M days prior to the sampling time s (in addition to the previous assumption of small $\lambda(s; Z)$ and survival distribution nearly 1) we have

$$\text{VE}_{\text{PI}}(s) = 1 - \exp\{\bar{\theta}(s)\} \frac{\Delta_1(s)}{\Delta_0}, \quad (3.7)$$

where $\exp\{\bar{\theta}(s)\} = \sum_{t=s-M+1}^s \exp\{\theta(t)\}/M$ and $\Delta_1(s) = M\pi_1(s)$ where $\pi_1(s)$ is the probability an infection in the vaccine arm that starts on $s - M + 1, \dots, s$ is detected on day s .

We additionally show that

$$\text{VE}_{\text{PVL}}(s) = 1 - \exp\{\bar{\theta}(s)\} \frac{\Delta_1(s)}{\Delta_0} \frac{v_1(s)}{v_0}, \quad (3.8)$$

where

$$v_1(s) = \frac{\sum_{t=s-M+1}^s \exp\{\theta(t)\} \sum_{d=s-t+1}^M p_1(d, t) v_1(s, t+1, d, t)}{\sum_{t=s-M+1}^s \exp\{\theta(t)\} \sum_{d=s-t+1}^M p_1(d, t)},$$

the weighted per day mean viral load for an infection detected at day s postvaccination.

3.3. Cox regression

Our above development for representing VE_{PI} and VE_{PVL} was for cross-sectional sampling. Trials may also do frequent, say daily or twice a week, sampling to detect infection, perhaps in a subset of participants. In this case, Cox regression can be used to estimate $\theta(s)$ and the frequent sampling after the detection of infection can be used to model the viral load trajectory and duration. Under time-constant vaccine efficacy and daily sampling, we can estimate $\Delta_0, \Delta_1, v_0, v_1$ with simple moment based estimates. With time-varying vaccine efficacy or less frequent sampling, we need to model the distribution of viral loads and account for data missing by design (if sampling is not daily). Such modeling can be complex but allows estimation of $\Delta_0, p_1(d, s), v_0$, and $v_1(s)$ based on the viral load data. Based on these, estimates of $VE_{PI}(s)$ and $VE_{PVL}(s)$ can be crafted using the equations (3.7) and (3.8). Possibilities for modeling include the bilinear mixed effects model used later to generate data in this article or models based on viral kinetics standard errors can be calculated using the delta-method or bootstrap.

4. ESTIMATION

To allow for multiple cross-sectional sampling times, let Y_{Zki} be the indicator that person i at visit k in arm Z is positive. Let s_{ki} be the time since vaccination, X_i be baseline covariates, and c_{ki} the calendar time since start of the study. We will use a binomial model with a log-link to model the probability of infection. This model has the advantage that it readily allows an estimate of vaccine efficacy as one minus a relative risk. One specification that can allow for waning efficacy is given as

$$P(Y_{Zk} = 1) = \exp[\phi_0(c_k) + X\beta + Z\{\phi_1 + \log(s_k)\phi_2\}]. \quad (4.9)$$

The term $\phi_0(c_k)$ allows for the risk of acquisition to depend on the calendar time of visit k, c_k . Possibilities include $\phi_0(c_k) = \phi_0$, a piecewise constant function, or possibly $\phi_{00} + \phi_{01}f(c_k)$, where $f(c)$ is a smoothed incidence rate for disease within the trial or from surveillance data. Functions of s_k other than the log, such as splines or polynomials, can be used to better approximate the true underlying curve. An estimate of vaccine efficacy on prevalent infection from (4.9) is

$$\widehat{VE}_{PI}(s) = 1 - \exp\left\{\hat{\phi}_1 + \hat{\phi}_2 \log(s)\right\}.$$

For viral load, we develop models conditional on infection, rather like a hurdle model (see Cragg, 1971). Analogous to (4.9), we specify a log-linear model conditional on covariates for the mean viral load on a random (length-biased sampled) day

$$v_Z(s_k) = E(V_{zk} | Y_{zk} = 1) = \exp[\omega_0 + X'\eta + Z\{\omega_1 + \log(s_k)\omega_2\}]. \quad (4.10)$$

The viral load V is often based on a log transformation. For settings where an infected person has a viral load below the limit of detection (LOD), one can use $LOD/2$ and then take the log.

The regression based VE_{PVL} at time s involves both (4.9) and (4.10)

$$\text{VE}_{\text{PVL}}(s) \approx 1 - \exp\{\phi_1 + \phi_2 \log(s)\} \times \exp\{\omega_1 + \omega_2 \log(s)\},$$

a blending of the time-varying vaccine effect on detecting infection under point sampling, times the time-varying effect on per day viral load, given infection. Under our model, the parameter estimates from (4.9) and (4.10) are independent. Thus, the delta-method could be used for testing and construction of confidence intervals. A simpler approach would be to bootstrap individuals.

4.1. Comments

- If vaccine efficacy is constant in the interval $(t - M, t)$, we set $(\omega_2, \phi_s) = (0, 0)$ and the regression estimators for a given point in time reduce to the simple estimators (2.2) and (2.3).
- One can always perform these analyses within subsets of subjects. For example, we might apply (4.9) to previously uninfected subjects as documented by serology, previous subclinical infection, or previous COVID-19 disease.
- While our focus has been on acquiring infection, irrespective of subsequent disease, one can also apply these methods to infection without subsequent symptoms. The impact of vaccine on proxies of transmission for these “silent” infections can be of interest and complement the proxies based on any infection.
- If vaccine efficacy did not vary with time, and the ratio Δ_1/Δ_0 were known or estimated from a different study, one could craft an estimate of the classic metric VE_I that ignores infection duration as

$$\widehat{\text{VE}}_I = 1 - (1 - \widehat{\text{VE}}_{\text{PI}}) \frac{\Delta_0}{\Delta_1}.$$

If there were no data to directly estimate Δ_1/Δ_0 , one might specify different plausible values of Δ_1/Δ_0 to obtain a range of VE_{PI} estimates.

- Given the connection between the Cox and point sampling models, one could design a trial where a subset of participants were subject to frequent (e.g., daily or twice-weekly) sampling and the rest subject to multiple point sampling. With time-varying vaccine effects, it can be complicated to equate the Cox and regression parameters, but for time-constant vaccine effects, it is relatively easy. We could estimate $\theta, \Delta_0, \Delta_1, \mu_0, \mu_1$ in the subset with frequent sampling and $\exp(\theta)\Delta_1/\Delta_0, \exp(\theta)\mu_1/\mu_0$ using point sampling. We could then combine the two estimates using inverse variance weights of the statistically independent estimates.

5. SIMULATIONS

5.1. Single point in time

In this section, we evaluate estimates at a single point in time postvaccination under a few different scenarios meant to roughly approximate COVID-19 vaccine trials. A large clinical trial was assumed with 15 000 per arm, and a total of 500 infection times were generated uniformly over the interval 0–200 with VE_I set at 0.00, 0.50, or 0.75. We assume that the viral load trajectory of an infected placebo volunteer follows a two stage model, with mean trajectory rising to six logs over the course of 4 days followed by a

Table 1. Simulated performance of the simple cross-sectional estimates of VE_{PI} and VE_{PVL} . The mean number of infections per arm is recorded along with the mean VEs and viral loads. The placebo arm has a mean peak viral load of six and a mean duration of 28 days for all scenarios. Each pair of rows reports the Monte Carlo means and (variances). Each row is based on 1000 simulated trials.

Input parameters					Estimated parameters					
VE_I	VE_{PI}	VE_{PVL}	Peak VL	Δ_1	# Positive		$\widehat{VE}_{PI}$	$\widehat{VE}_{PVL}$	$\bar{V}_0$	$\bar{V}_1$
					Vaccine	Placebo				
0.00	0.00	0.00	6	28	34.3 (32.1)	33.9 (35.5)	-0.047 (0.071)	-0.067 (0.101)	3.04 (0.087)	3.07 (0.098)
0.50	0.75	0.75	6	14	11.1 (11.5)	45.5 (39.7)	0.750 (0.007)	0.740 (0.011)	3.04 (0.066)	3.13 (0.298)
0.50	0.50	0.67	4	28	22.7 (22.4)	45.5 (39.7)	0.490 (0.018)	0.654 (0.011)	3.04 (0.066)	2.04 (0.067)
0.50	0.75	0.83	4	14	11.1 (11.5)	45.5 (39.7)	0.750 (0.007)	0.826 (0.005)	3.04 (0.066)	2.09 (0.137)
0.75	0.88	0.88	6	14	6.7 (6.5)	54.5 (50.9)	0.874 (0.002)	0.868 (0.003)	3.04 (0.059)	3.15 (0.548)
0.75	0.75	0.83	4	28	13.8 (12.7)	54.5 (50.9)	0.742 (0.006)	0.825 (0.003)	3.04 (0.059)	2.05 (0.111)
0.75	0.88	0.92	4	14	6.7 (6.5)	54.5 (50.9)	0.874 (0.002)	0.912 (0.001)	3.04 (0.059)	2.11 (0.255)

linear decline to zero over a mean of 28 days (see [To and others, 2020](#)). Vaccine infections were similar but with a peak viral load of six or four logs and a mean time to zero of 14 or 28 days. Each infected volunteer drew random deviations from the mean linear rise and mean time to zero which were Gaussian with standard deviations 0.125, 2.00, respectively with correlation 0.50. Each day the measured viral load for an infected individual was given by the mean viral load plus Gaussian error with a standard deviation of 0.10. Infection ended when the simulated viral load was first less than zero. We approximated the true Δ_0 , Δ_1 , μ_0 , and μ_1 by simulation of 100 000 infection episodes per arm which allowed us to approximate the true VE_{PI} and VE_{PVL} .

A cross-sectional study was conducted on day 100. Any volunteer with a detectable viral load on day 100 was tallied and their viral load recorded. Table 1 presents the results. The estimates are close to the true values. The average $\widehat{VE}_{PI}$ increases as the vaccine induced duration decreases and the average $\widehat{VE}_{PVL}$ increases as the vaccine induced peak decreases. The placebo mean viral load $\bar{V}_0$ is the same over all scenarios—as the placebo generation model is the same. The vaccine mean viral load $\bar{V}_1$ decreases with a lower peak viral load but stays the same if the peak is unchanged but the duration is changed. The variances of $\widehat{VE}_{PI}$ and $\widehat{VE}_{PVL}$ decrease with increasing VE. In contrast, for fixed peak VL and duration, the variance of $\bar{V}_1$ increases with increasing VE as there are fewer events with which to estimate μ_Z . For fixed VE_I , the variance of $\bar{V}_1$ is smallest for the scenarios with a peak viral load of 4 and duration of 28 days.

We selected the last simulated data set from the last row of Table 1 to illustrate an analysis. Figure 3 provides the generated viral load trajectories of the infected volunteers and projects the day 100 viral load onto the right axis. For this setting, the true $VE_I = 0.75$ and the vaccine reduces peak viral load by 2 logs and cuts the duration in half. There were a total of six infections on vaccine and 43 on placebo for a cross-sectional estimate of $\widehat{VE}_{PI} = 1 - \frac{6/15000}{43/15000} = 0.86$. The mean viral loads on the two arms were 2.06 and 3.07. Thus, $\widehat{VE}_{PVL} = 1 - \frac{6}{43} \frac{2.06}{3.07} = 0.91$. The variances of the vaccine and placebo viral loads were 0.33

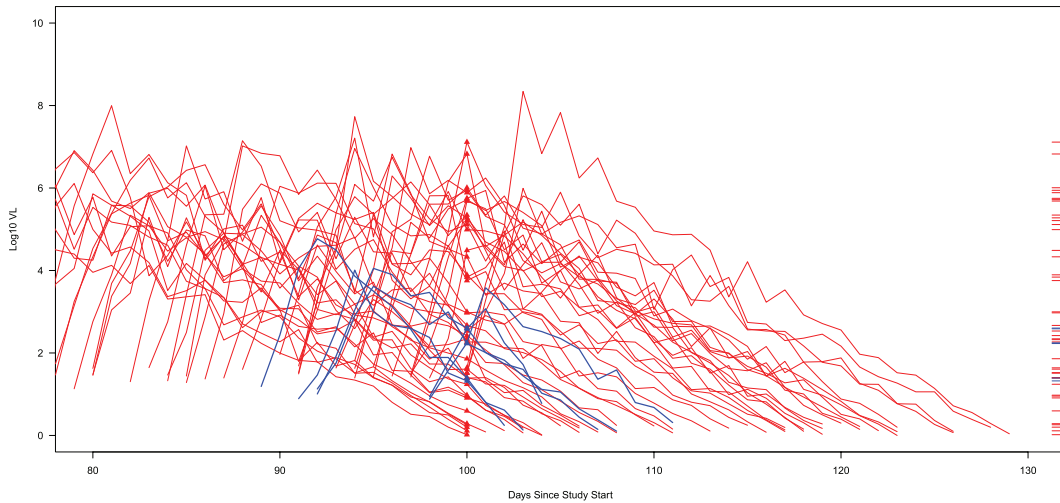


Fig. 3. Simulated data set for a cross-sectional study with true $VE_I = 0.75$, where vaccine reduces the peak viral load by 2 logs and the duration of infection by 14 days. Samples are drawn at day 100 to identify infections and viral loads, thus only the ticks on the far right are observed. Blue denotes vaccine and red placebo.

and 4.47, respectively, though the standard errors of the mean were more similar at $\sqrt{0.33/6} = 0.23$ and $\sqrt{4.47/43} = 0.32$. We use the percentile bootstrap to form confidence intervals. With 10 000 bootstrap samples the 95% confidence intervals are (0.71–0.96) and (0.80–0.95), respectively, for VE_{PI} and VE_{PVL} .

In Appendix A, we examine the sensitivity of the simulation results to the assumption of a constant attack rate before the day of point sampling. Given our durations are at most about 30 days, we varied the attack rate to either increase 3-fold or reduce by a factor of 3 from day 70 to day 100. Based on the pandemic in the United States, a factor of 3 seemed a plausible worst case scenario. The impact on estimates of VE_{PI} and VE_{PVL} was modest with less than a 10% bias under the least favorable setting. The mean viral loads increased about 15% under the increasing attack rate setting and decreased about 10% under the decreasing attack rate setting.

5.2. Multiple points in time

Our next set of simulations approximate the follow-up of the COVE trial of the mRNA-1273 vaccine with a smaller sample size and higher attack rate. In COVE, all subjects were sampled for infection at day 29 postvaccination and then again during the participant decision visit (PDV), where volunteers were unblinded and placebo recipients offered vaccine. The PDVs occurred in December through March. Thus, all participants were swabbed at a fixed time relative to vaccination and then variable times when placebo volunteers were offered vaccination with mRNA-1273.

We specify 5000 per arm with a cross-sectional sample at day 50 and then another sample selected randomly over the interval 100–160. For the placebo arm we generate infections according to an exponential model with parameter $\lambda_0 = -\log(0.92)/160$ so that about 8% of the subjects should be infected by day 160. We then generate vaccine infections by a Weibull model with $\lambda_1(s) = \lambda_1 \exp(\alpha \log(s))$ with $\log(\lambda_1/\lambda_0) = -3.72$ and $\alpha = 0.597$. It follows that

$$VE_I(s) = 1 - \exp\{-3.72 + 0.597 \log(s)\}$$

Table 2. Simulated performance of the regression estimates of VE_{PI} and VE_{PVL} for a study with two point samples. The first sample was 50 days postrandomization and the second uniformly distributed over days 100–160. The mean number of infections per arm is recorded along with the mean estimated VEs at 50 and 130 days postvaccination. The placebo arm has a mean peak viral load of 6 and a mean duration of 28 days for all scenarios. Pairs of rows denote the Monte Carlo means (variances). Each row is based on 1000 simulated trials

Input parameters						Estimated parameters					
Peak	Δ_1	VE_{PI}		VE_{PVL}		No. of positive		$\widehat{VE}_{PI}(t)$		$\widehat{VE}_{PVL}(t)$	
VL		$t = 50$	$t = 130$	$t = 50$	$t = 130$	Vaccine	Placebo	$t = 50$	$t = 130$	$t = 50$	$t = 130$
6	28	0.794	0.585	0.806	0.592	43.4 (44.7)	138.0 (144.4)	0.786 (0.002)	0.582 (0.005)	0.777 (0.003)	0.566 (0.006)
4	28	0.794	0.585	0.871	0.727	43.4 (44.7)	138.0 (144.4)	0.786 (0.002)	0.582 (0.005)	0.851 (0.001)	0.710 (0.003)
6	14	0.885	0.785	0.890	0.787	22.6 (23.4)	138.0 (144.4)	0.882 (0.001)	0.789 (0.002)	0.878 (0.001)	0.781 (0.003)
4	14	0.885	0.785	0.926	0.858	22.6 (23.4)	138.0 (144.4)	0.882 (0.001)	0.789 (0.002)	0.918 (0.001)	0.854 (0.001)

with $VE_I(50) = 0.75$ and $VE_I(160) = 0.50$. Infection times are rounded to the nearest day. The duration of infection and distribution of viral loads, among the infected are as given in the previous section and do not depend on time since vaccination.

With this structure and under the rare disease assumption, we have

$$P(Y_Z(s) = 1|Z) \approx \exp\{\omega_0 + Z(\phi_0 + \phi_1 \log(s))\},$$

where $\phi_1 = \alpha$, $\omega_0 = \log(\lambda_0 \Delta_0)$ and $\phi_0 = \log\{\lambda_1 \Delta_1 / (\lambda_0 \Delta_0)\}$. For the viral load conditional on infection, we have no dependence on time and

$$v_Z = E(V_{ik}|Y_{ik} = 1) = \exp(\omega_0 + \omega_1 Z_i),$$

so $\omega_0 = \log(v_0)$ and $\omega_1 = \log(v_1) - \log(v_0)$.

We estimate the parameters of interest using

$$\widehat{VE}_{PI}(s) = 1 - \exp\{\hat{\phi}_0 + \hat{\phi}_1 \log(s)\}$$

and

$$\widehat{VE}_{PVL}(s) = 1 - \exp\{\hat{\phi}_0 + \hat{\phi}_1 \log(s) + \hat{\omega}_1\}$$

at $s = 50, 130$. For each scenario, we simulated 1000 clinical trials. Table 5.2 reports the results. We see that for all scenarios the parameters are approximately unbiased, indicating good recovery even with approximately 8% of the placebo group infected. We then evaluated how to combine Cox and point sampling regression estimates under time-constant vaccine efficacy. We generated data for the last scenario of Table 2, except infection times for the vaccine group were exponential with $\lambda_1/\lambda_0 = 1/3$ so that $VE_I = 1 - 1/3 = 2/3$. Out of the 5000 randomized per arm, we randomly selected 1000 (500) subjects to have daily sampling and the remainder to have two point samples as described above. We calculated the

Monte Carlo variances for the Cox and regression based point sample estimators for the VE_{PL} and VE_{PVL} parameters. The ratio of Cox divided by point sample variance were 0.43 (0.99) for VE_{PI} and 0.35 (0.81) for VE_{PVL} . Thus, the subset of 500 with daily sampling and $80\,000 = 160 \times 500$ swabs has about the same information content as the 4500 who are sampled twice with 9000 swabs. While cross-sectional sampling is much more “swab” efficient, an advantage of the daily swabbing is that the entire viral load distribution can be estimated rather than just the VE_{PI} and VE_{PVL} parameters, which is all that point sampling can do. Of course, this calculation depends on the expected duration and attack rate used in this simulation. As such, it serves more to provide a framework for design and less as a universal rule-of-thumb.

6. EXAMPLE

The Cove trial randomized approximately 30 000 volunteers equally to two doses of the vaccine mRNA-1273 or placebo delivered 28 days apart. The trial was designed to achieve 150 cases of symptomatic COVID-19 disease which was actively monitored (see [Moderna, 2020](#)). At the second vaccination visit, serology and PCR sampling were performed to identify subjects who were both serologically and PCR negative. The trial reported high efficacy on the primary endpoint of symptomatic disease at the first interim analysis and quickly applied for an Emergency Use Authorization from the Food and Drug Administration. Among baseline seronegative volunteers, the reported PCR positive rates at Day 28 for asymptomatic volunteers, were 14/14 134 and 38/14 073, respectively for the vaccine and placebo arms (see [Moderna, 2020](#)). Using (2.2), we estimate VE_{PI} as 0.63. A 95% bootstrap confidence interval based on 10 000 bootstrap samples is 0.34–0.82). Thus using a cross-sectional design provides useful information about the efficacy of 1 dose of the mRNA-1273 vaccine on silent infections that do not progress to symptomatic disease.

While the mean duration of silent infections is not yet known for unvaccinated nor mRNA-1273 vaccinated volunteers, we can specify some ratios Δ_1/Δ_0 to give a range of plausible estimates of the traditional VE_I metric. We illustrate by postulating that Δ_1/Δ_0 ranges from 0.50 to 1.00. That is, the vaccine cuts the mean duration in half, leaves it unchanged, and everything in between. This results in associated VE_I estimates ranging from $1 - (14/14\,134)/(38/14\,073) \times 2 = 0.27$ to 0.63.

7. DISCUSSION

Necessity can be the mother of invention. This work was motivated by the necessity of interpreting infection counts and viral loads from a single cross-sectional sample of asymptomatic trial volunteers. We demonstrate that such samples can be used to estimate two meaningful metrics; the vaccine efficacy on prevalent infection and the vaccine efficacy on prevalent viral load. Pleasingly, these metrics should be better proxies for the effect of vaccine on transmission than traditional metrics which focus on the effect of a vaccine on individual infection and individual mean viral load irrespective of duration. For transmission, what matters is the transmission potential induced by the vaccine on any given day and the new metrics naturally capture this aspect. We recommend that future studies, whether cross-sectional or frequent longitudinal sampling, use these metrics to help describe the manifold effects of a vaccine.

In general, frequent longitudinal sampling will be much more efficient per person than single point sampling as many more infections will be captured. Furthermore, the entire viral load distribution can be estimated. In some settings, cross-sectional sampling will not be feasible as too few cases will be accrued. Nonetheless, for large studies where longitudinal sampling is not feasible and there is a high “capture” rate of infections, cross-sectional sampling can have adequate power to test for non-null effects of VE_{PI} and VE_{PVL} . For example with 90 infection cases from point sample(s), we have 90% power to detect a VE_{PI} of 0.50. Thus if longitudinal sampling for infection is not logistically feasible, cross-sectional sampling should be considered as a practical design that allows estimation of the effect of vaccine on proxies for

transmission. To gain efficiency, sampling times could be chosen when the attack rate is relatively high, for example, with seasonal diseases or if community surveillance indicates a surge.

Future work could explore how to augment a cross-sectional study with subsequent daily sampling of infected volunteers, known as a prevalent cohort study (Brookmeyer and others, 1987; Degruittola and others, 1991). Here, serial viral loads and time to infection cessation would be collected. While our results appear robust to the constant attack rate assumption for COVID-19, for other diseases with varying attack rates and longer durations, this may be less plausible (e.g., HIV and AIDS). Work on how to weaken this assumption could also be explored. Another generalization would be to apply these methods to trials where the placebo group has been vaccinated (Follmann and others, 2021). Finally, while our work focused on the binary outcome of infection, in some settings one can subdivide infections into those that result in disease versus those that are purely asymptomatic. Such subdivision could lead to a multinomial regression model for the two competing events of infection that progresses to diseases versus infection without disease.

SUPPLEMENTARY MATERIAL

Supplementary material is available at <http://biostatistics.oxfordjournals.org>.

ACKNOWLEDGMENTS

We thank Elizabeth Brown, Peter Gilbert, Holly Janes, Jing Qin, and Stephanie Schrag for helpful comments. Code to reproduce the example is available at <https://github.com/follmand/Vaccine-Efficacy-At-A-Point-In-Time>.

Conflict of Interest: None declared.

APPENDIX

A. SIMULATIONS WITH CHANGING ATTACK RATES

Our development has been for the setting where the attack rate is assumed roughly constant for the period prior to the cross-sectional sample. The incidence of COVID-19 has changed over the course of the pandemic. To assess the sensitivity of our simulation results to the constant attack assumption, we did some simulations. Our duration distribution has a maximum duration of about 30 days. In the United States, over the course of the pandemic the largest change in incidence over any 30-day period has been about a tripling. We thus evaluate two scenarios. Under the increasing scenario, the attack rate at day 100 was 3 times the attack rate at day 70. Under the decreasing scenario, the attack rate at day 100 was one-third the attack rate at day 70. For each simulated study, we generate 70 infections over the period from day 70 to day 100 under the increasing or decreasing scenario and sampled PCR+ infections at day 100.

Table A1 below reports the results. We do see some bias in the sampled mean viral loads. Under the constant attack rate scenario of Table 1, and with peak VL and duration of 6 and 28, respectively, the mean sampled viral loads were about 3.0. With a peak VL of 4, the mean viral load was about 2.0 (regardless of duration). With an increasing attack rate these mean viral loads increase to about 3.4 and 2.3, respectively or an increase of about 15%. With a decreasing attack rate, the mean viral loads decrease to about 2.7 and 1.8 or about at 10% decrease.

Under both scenarios, VE_{PI} appeared unbiased if the vaccine had no effect on duration. Under the increasing (decreasing) attack rate scenario, VE_{PI} was slightly biased downward (upward) if the vaccine reduced the duration. The bias was less than 10%, often less than 5%. A similar pattern but with less bias was observed for VE_{PVL} .

Table A1. *Simulated performance of the simple cross-sectional estimates of VE_{PI} and VE_{PVL} . The mean number of infections per arm is recorded along with the mean VEs and viral loads. Each row is based on 1000 simulated trials. The placebo arm has a mean peak viral load of 6 and a mean duration of 28 days for all scenarios*

Inputs					Performance					
VE _I	VE _{PI}	VE _{PVL}	Peak VL	Infection Duration	No. of positives		$\widehat{VE}_{PI}$	$\widehat{VE}_{PVL}$	$\overline{V}_0$	$\overline{V}_1$
					Vaccine	Placebo				
Attack rate at Day 100 is 3 times that of Day 70										
0.00	0.00	0.00	6	28	33.30	33.20	−0.033	−0.044	3.42	3.42
0.50	0.75	0.75	6	14	13.50	44.40	0.689	0.702	3.41	3.25
0.50	0.50	0.67	4	28	22.30	44.40	0.485	0.651	3.41	2.29
0.50	0.75	0.83	4	14	13.50	44.40	0.689	0.800	3.41	2.18
0.75	0.88	0.88	6	14	8.01	53.20	0.847	0.855	3.42	3.20
0.75	0.75	0.83	4	28	13.30	53.20	0.744	0.829	3.42	2.28
0.75	0.88	0.92	4	14	8.01	53.20	0.847	0.903	3.42	2.14
Attack rate at Day 100 is 1/3 times that of Day 70										
0.00	0.00	0.00	6	28	30.70	30.70	−0.032	−0.046	2.70	2.69
0.50	0.75	0.75	6	14	7.83	40.90	0.804	0.782	2.69	2.94
0.50	0.50	0.67	4	28	20.60	40.90	0.483	0.648	2.69	1.81
0.50	0.75	0.83	4	14	7.83	40.90	0.804	0.854	2.69	1.96
0.75	0.88	0.88	6	14	4.70	48.90	0.902	0.893	2.68	2.91
0.75	0.75	0.83	4	28	12.30	48.90	0.743	0.824	2.68	1.82
0.75	0.88	0.92	4	14	4.70	48.90	0.902	0.928	2.68	1.95

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[Received February 4, 2021; revised February 2, 2022; accepted for publication February 6, 2022]